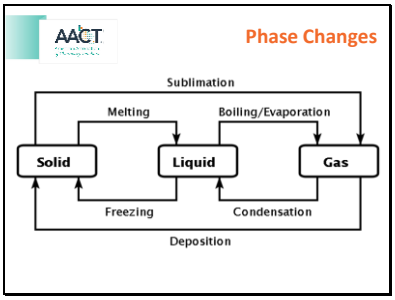
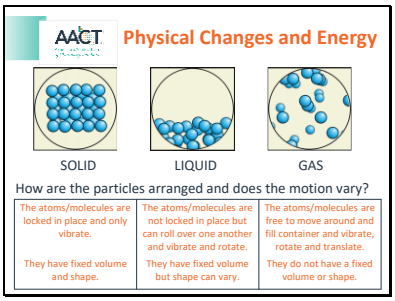
CHEMISTRY BASICS AND MEASUREMENTS – DAY 5
GUIDED NOTES AND PRACTICE PROBLEMS

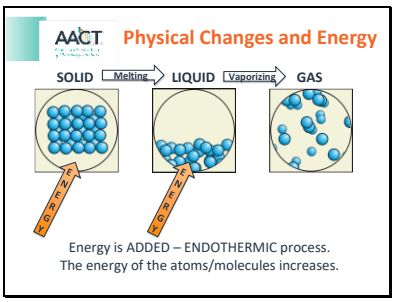


What is a "change of state"?

Explain how the properties of a SOLID and LIQUID and GAS differ in terms of volume, shape and motion.

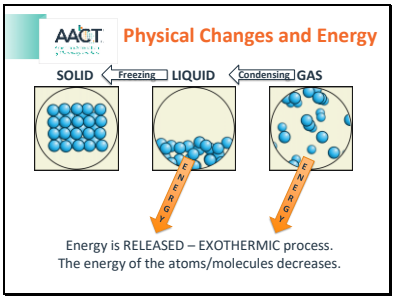


Explain why melting and boiling (vaporizing) are endothermic changes:



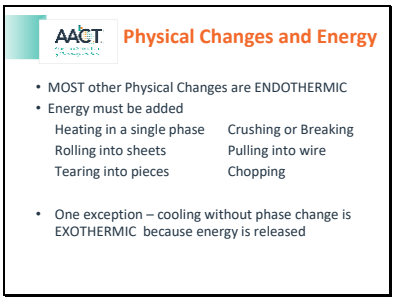
How does the energy level change as you move from solid to liquid to gas?

Explain why condensing and freezing (vaporizing) are exothermic changes:



How does the energy level change as you move from gas to liquid to solid?

Explain why crushing, rolling out, tearing, and heating are exothermic changes:



Chemical Changes and Energy

EXOTHERMIC REACTIONS	ENDOTHERMIC REACTIONS

• Reaction produces energy
• Energy is released to and absorbed by the surroundings
• Evidence:
  • Feels hot
  • Reaction vessel temperature increases
  • You see fire or light
	• Reaction absorbs energy
	• Energy flows from the surroundings to the reaction
	• Electric current applied
	• Evidence
	  • Feels cold
	  • Reaction vessel temperature decreases

Exothermic reactions _____ energy.

What will serve as evidence of an exothermic reaction?

- _____
- _____
- _____

Endothermic reactions _____ energy.

What will serve as evidence of an endothermic reaction?

- _____
- _____

Exo- or Endothermic?

Step 1: Is it a physical or chemical Change??
Step 2: Think about the energy flow.

Physical Changes (Phase Changes/Reshaping)	Chemical Changes (Reactions)
Think about energy flow – Heat Added – ENDO Heat Released – EXO	Think about energy flow – Heat Absorbed – ENDO Heat Produced – EXO
Think about the energy and movement of particles.	Is there an unexpected temperature change?

How do you decide if a change is physical or chemical?

Endo- or Exothermic?

1. Ammonium chloride dissolves in water. The test tube feels hot to the touch.
2. Hydrochloric acid is added to zinc and produces hydrogen gas bubbles. The test tube feels hot.
3. Calcium reacts with water, producing hydrogen bubbles. The temperature increases.

1 – Endothermic or exothermic?

2 – Endothermic or exothermic?

3 – Endothermic or exothermic?

Endo- or Exothermic?

4. Calcium chloride dissolves in water. The bag feels warm to the touch.
5. Ammonium nitrate dissolves in water. The bag feels cold to the touch.
6. Hydrochloric acid is added to baking soda. Gas bubbles are produced and the test tube feels cold.

4 – Endothermic or exothermic?

5 – Endothermic or exothermic?

6 – Endothermic or exothermic?

Endo- or Exothermic?

7. Aluminum metal melts when heated to 660°C.
8. Methane gas combusts with oxygen forming carbon dioxide and water vapor and a large amount of heat and light.
9. Water vapor condenses when it is cooled to below 660°C.
10. Copper metal is hammered into a sheet.

7 – Endothermic or exothermic?

8 – Endothermic or exothermic?

9 – Endothermic or exothermic?

10 – Endothermic or exothermic?

Energy Changes in Chemistry

Complete the following tasks before our next class:

- Complete the activity sheet related to today's topics if we did not complete it in class.
- Review class notes about lab safety and equipment, and matter, classification of matter, and exothermic and endothermic changes.

Notes about what to complete before the next class: